Whose Boat Does it Float?
Improving Personalization in Preference Tuning via Inferred User Personas

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Abstract

LLMs are aligned to follow input instructions by learning which of two responses users prefer for a prompt. However, such preference data do not convey *why* users prefer responses that are chosen or rejected, so LLMs trained on these datasets cannot tailor responses to varied user needs. To surface these parameters of personalization, we apply *abductive reasoning* to preference data, inferring needs and interests of users, i.e., personas, that may prefer either response. We test this idea in two steps: **Persona Inference (PI)**—abductively inferring personas of users who prefer chosen or rejected outputs—and **Persona Tailoring (PT)**—training models to tailor outputs to personas from PI. We show: 1) LLMs infer personas accurately explaining why different users may prefer *both* chosen or rejected outputs; 2) Training on preference data augmented with PI personas via PT boosts personalization and generalizes to supporting user-written personas; and 3) Rejected response personas form harder personalization evaluations, showing PT better aids users with uncommon preferences versus typical alignment methods. We argue for an abductive view of preferences for personalization, asking not only which response is better but when, why, and for whom.[1]

1 Every Preference Happens for a Reason

Current methods for aligning large language models (LLMs) predominantly use preference data (Ji et al., 2023), created by finding: for a given prompt, which of two outputs do users prefer? LLMs are then trained on this data via preference tuning methods like direct preference optimization (Rafailov et al., 2024, DPO), learning to give outputs like the majority *chosen* response and unlike the *rejected* one. This improves LLM outputs in dialogue (Köpf et al., 2024), question answering (Fan et al., 2019, QA), and summarization (Völske et al., 2017).

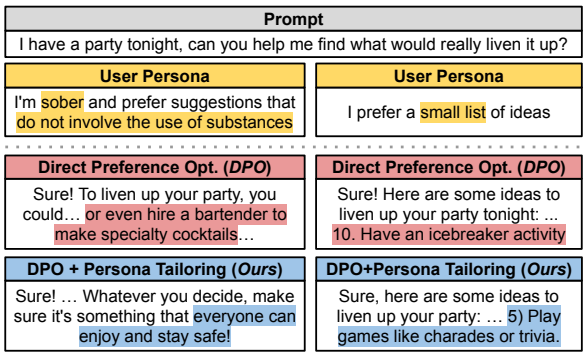


Figure 1: Training methods with typical preference datasets like DPO cannot fully cater to a user's specified personas. To overcome this, we train models on preference data augmented with LLM-inferred personas, which we call persona tailoring.

While preference datasets are valuable, they assume chosen responses are universally better, failing to consider *why* users prefer responses (Joshi et al., 2025). In reality, some users genuinely prefer rejected outputs, even if their reasons are less common. In Figure 2 (left), most users prefer the chosen output for taking brownies to a cake sale. While users valuing simplicity may prefer this direct response, practical users may favor the rejected one, as it also gives packaging logistics (Figure 2, middle). Since users prefer responses for varied reasons (Kirk et al., 2024), models trained on preference data should personalize outputs to meet these specific, individual needs (Salemi et al., 2024).

A common, interpretable method for personalization involves users specifying their needs and interests as an extra input (e.g., "I like short answers")—forming a system prompt **persona** to tailor model responses (Zhang et al., 2024c). However, current preference dataset formats lack personas explaining why users prefer responses, overlooking signals for training personalized models (Lee et al., 2024).

The only way for models trained on existing preference dataset formats to use a persona is by adding it to the inference prompt, hoping the model treats it as an extra instruction to follow (Deb et al., 2022). However, standard preference tuning methods like

[1]Code and datasets are available at: https://github.com/Pinafore/alignment-personalization

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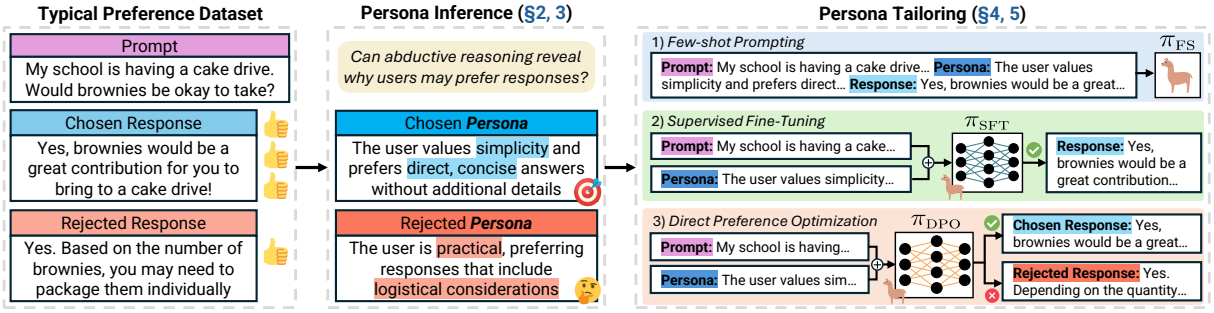


Figure 2: **Overview of this paper.** Preference data has a prompt, chosen response, and rejected response (left). Most users prefer the chosen response, but there are valid reasons and personas of users that may prefer either response, which we uncover via abductive reasoning in PERSONA INFERENCE (PI, middle). We then study PERSONA TAILORING (PT) to use the personas for data augmentation in few-shot prompts, fine-tuning, and direct preference optimization—enhancing personalization (right).

DPO cannot naturally adapt to personas in this way. In Fig 1, DPO tells a sober user to hire a bartender (left), while a user desiring a small list gets 10 ideas (right); such responses are unhelpful, inappropriate, and do not tailor to the users' specified personas.

As models' poor personalization stems from a lack of training on personas, we propose **abductive reasoning** (Peirce, 1974) to augment preference training data with LLM-inferred personas. Abduction infers hidden contexts that explain a given outcome (Zhao et al., 2024). Similarly, we adopt this reasoning to infer (i.e., adduce) hidden needs, interests, and traits of users, i.e., personas (Tseng et al., 2024), that explain why chosen and rejected outputs may be preferred. If LLMs surface valid personas of users who may prefer chosen or rejected responses, we can attach them to preference datasets and train models that tailor to user-specified needs, enhancing model personalization (Figure 1, blue).

We segment this idea into a two-task sequence: **1) Persona Inference (PI**, §2**):** adducing personas for preference data responses (Figure 2, middle); and **2) Persona Tailoring (PT**, §4**):** producing outputs tailored to personas from PI (Figure 2, right).

We first test LLMs in PI on dialogue (Bai et al., 2022a), QA (Ji et al., 2024), and education (Balepur et al., 2024c) preference data. LLM personas accurately convey different users who could prefer chosen *or* rejected outputs; LLaMA-405B has 91% accuracy, judged by GPT-4o with 90% human agreement (§3.1). Further, chosen and rejected response personas are often judged as tied in quality (§3.2), and humans rate rejected ones as plausible but applying to fewer users (§3.3). Thus, users may prefer rejected outputs for uncommon but valid reasons. Personas are also a useful content analysis tool to find differences in chosen and rejected responses; in BeaverTails (Ji et al., 2024), chosen response personas describe users who are "meticulous", while rejected ones describe "direct" users, showing these labelers may prefer verbosity (Zheng et al., 2024a).

As PI yields accurate personas, we augment preference data with PI personas using LLaMA-405B. We then train LLaMA-8B on this new data for the reverse task of PT: using prompts and inferred personas as inputs to give tailored responses (Figure 2, right). We test three strategies: prompting (Brown et al., 2020, PT_{FS}), fine-tuning (Chung et al., 2024, PT_{SFT}), and DPO (Rafailov et al., 2024, PT_{DPO}).

Each generation strategy largely boosts personalization when using PT (§5.1), with PT_{DPO} being the strongest (§5.2). Further, DPO fine-tuned on preference datasets without personas cannot always tailor to personas during inference; notably, PT_{DPO} is judged as much stronger than DPO on uncommon but still valid needs in the personas from rejected responses (**66% average improvement in personalization**), showing rejected responses can form valuable, harder evaluations for personalization (§5.3). Finally, eight users author 144 diverse personas and rate PT_{DPO} and DPO responses for these personas (§5.4). The same users find our PT_{DPO} method personalizes more effectively to their written needs, showing models trained on realistic, LLM-inferred personas can generalize to real user-specific needs.

We argue for an abductive view of preferences, capturing not only which outputs users prefer but *which users* and *for what reasons*. In doing so, we can find valid user needs that may be overlooked in rejected responses (§3) and ensure LLMs support them (§5.3). It also improves personalization (§5), such as augmentation via PI. Our contributions are: **1)** We study abductive reasoning on preference data via persona inference (PI) to show LLMs can infer why users may prefer chosen and rejected outputs. **2)** We release persona-augmented question answering, dialogue, and education preference datasets.[1] **3)** Persona tailoring—prompting and training on

persona-augmented data—effectively personalizes to needs inferred by PI and specified by real users.

2 Persona Inference Setup

The first step in our personalization method is inferring why users may prefer each response in standard preference data (Fig 2, middle). We use abduction (Peirce, 1974)—which explains outcomes—to infer persona-based explanations for why responses may be preferred via **PERSONA INFERENCE (PI)**:

• **PI**$(p, r_1, r_2) \rightarrow \mathcal{P}_1$: For prompt p and responses r_1 and r_2, the LLM gives a persona $\mathcal{P}_1$ such that a user described by $\mathcal{P}_1$ would prefer r_1 over r_2. If r_1 is the chosen response and r_2 is the rejected one, $\mathcal{P}_1$ will describe a user preferring the chosen response, and vice versa if r_1 is the rejected response.

Following Chen et al. (2024a), $\mathcal{P}_1$ is closest to a demographic persona, but we infer broad traits—like information needs, interests, or personalities—rather than protected attributes (e.g., race) to curb stereotyping (Kantharuban et al., 2024). Our personas include no other constraints. All $\mathcal{P}_1$ follow the format: "The user is [attribute] and prefers [explanation of preference]" for parsing. This section gives our PI models (§2.1) and datasets (§2.2).

2.1 Models

We test nine LLMs in PI: **Claude** (Anthropic, 2023, Sonnet, Haiku, Opus), **GPT** (Achiam et al., 2023, 3.5, 4, 4o), and **L**LaMA-3.1 Instruct (Dubey et al., 2024, 8B, 70B, 405B). We use 5-shot prompts in the format below, where highlights are generations:

Prompt 2.1: Few-Shot Persona Inference Prompt

```
Prompt: p
Chosen Response: r1
Rejected Response: r2
Persona: P1
```

We append text to Prompt 2.1 asking for a "short, one-sentence description of the user's preference". We also specify the persona must have "high-level characteristics" to avoid stereotypes and not have exact phrases in the prompts or responses to avoid the trivial solution of repeating r_1 for $\mathcal{P}_1$ (§3.3).

2.2 Datasets

PI requires preference data with an input prompt p and responses $\mathcal{R} = [r_1, r_2]$. For a thorough evaluation, we use four datasets in question answering (QA), dialogue, and education—three domains that benefit from personalization (Zhang et al., 2024c):

1) BeaverTails (Ji et al., 2024) has advice queries p and candidate answers $\mathcal{R}$ on 14 harm categories including politics, privacy, and unethical behavior.

2) Stanford Human Preferences (Ethayarajh et al., 2022, **SHP**) has Reddit post questions p on r/ask or advice forums with user-written answers $\mathcal{R}$.

3) Anthropic HHH (Bai et al., 2022a) has human inputs p and assistant responses $\mathcal{R}$ from real dialogues. We use single-turn dialogues for simplicity.

4) Mnemonic (Balepur et al., 2024c) has vocabulary terms p and keyword mnemonics $\mathcal{R}$. Mnemonics are study aids that help users learn p's meaning.

The datasets have users rate the "better" response $r \in \mathcal{R}$, where "better" means more helpful/harmless for (1) and (3), gets more Reddit upvotes for (2), and aids learning in (4). The overall better r is chosen (r_C) and the other is rejected (r_R). For each entry, we alter the inputs r_1 and r_2 in PI to get chosen personas $\mathcal{P}_\text{C} = \text{PI}(p, r_1 = r_\text{C}, r_2 = r_\text{R})$ and rejected personas[2] $\mathcal{P}_\text{R} = \text{PI}(p, r_1 = r_\text{R}, r_2 = r_\text{C})$ for r_C and r_R. As r_R is preferred less often, we assume $\mathcal{P}_\text{R}$ has less common/popular needs (§3.3).

In BeaverTails and SHP, some responses r_R are deliberately low-quality, with harmful or inaccurate text (Liu et al., 2024b). These are out-of-scope, as all $\mathcal{P}_\text{R}$ are unsafe[3] and we do not want models to tailor to them (§9). Thus, we use the data split labeled safe in BeaverTails and outputs with 10+ upvotes in SHP. Post-filtering, we sample 300 entries in each dataset to form 600 PI inputs (Appendix A.1).

3 Evaluating LLM-Inferred Personas

We first verify LLM persona quality before using them to train personalized models (§4). As many personas can explain the same preference,[4] we lack ground truth. Instead, we verify that personas *accurately* explain why users may prefer preference data responses (§3.1)—the goal of PI (§2) and a common abduction metric (Balepur et al., 2025a)—with GPT-4o (90% human agreement). We then further study personas, showing chosen and rejected personas are similarly valid needs (§3.2, §3.3) and reveal preference dataset trends (§3.4). Thus, PI yields high-quality personas we can use in PT (§4).

3.1 LLMs Accurately Infer Personas

We first verify each LLM persona $\mathcal{P}_1$ fulfills abduction's goal: accurately justifying why its response

[2]We use "rejected" for brevity, not to imply they are worse.

[3]Such as "the user is unethical" or "likes misleading users".

[4]For the prompt "What should I eat tonight?" with responses "Pizza!" and "Go eat steak", a user may prefer the former if they like vegetarian options, enthusiasm, or brevity.

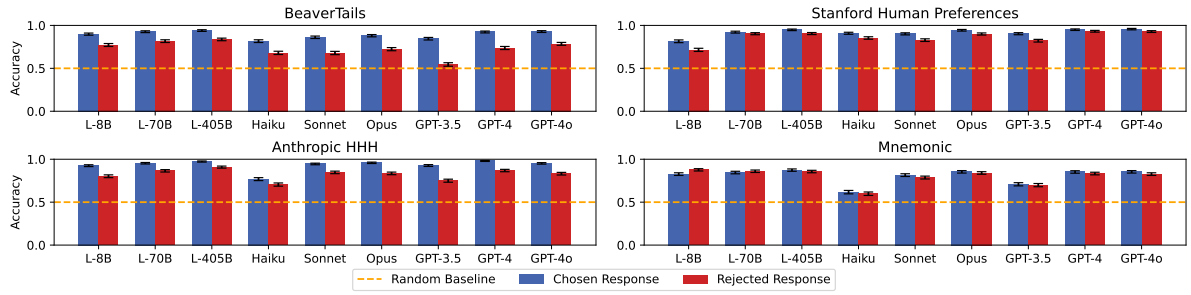


Figure 3: GPT-4o judgments on if LLM personas accurately infer users who prefer chosen/rejected responses. Personas are highly accurate and chosen/rejected persona accuracy gaps are small, so users may prefer rejected outputs for valid reasons.

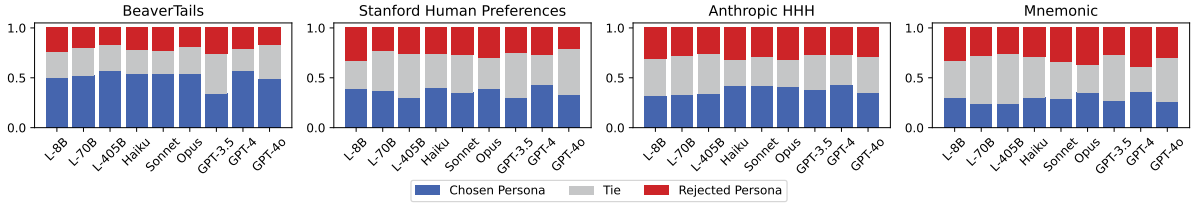


Figure 4: Persona quality comparison. Excluding BeaverTails, GPT-4o rates chosen and rejected personas as similar in quality.

r_1 may be favored over r_2 (§2). GPT-4o—a judge with 90% agreement with three Ph.D. students (Appendix A.4)—evaluates this, judging if a user described by $\mathcal{P}_1$ would prefer response r_1 or r_2 for the prompt p. A chosen persona $\mathcal{P}_\mathrm{C}$ is **accurate** if GPT-4o selects r_C over r_R (and vice versa for $\mathcal{P}_\mathrm{R}$).

LLMs infer accurate $\mathcal{P}_\mathrm{C}$ and $\mathcal{P}_\mathrm{R}$; the judge often picks the intended output (Figure 3). $\mathcal{P}_\mathrm{R}$ is usually less accurate, consistent with work showing LLMs struggle to justify incorrect answers (Balepur et al., 2024a). However, some models show small gaps in accuracy (0.06 for L-405B), so while $\mathcal{P}_\mathrm{R}$ is harder to infer, it can still reveal plausible needs of users. Finally, accuracy on Mnemonic is lowest, as LLMs must infer why outputs aid learning, which is likely harder than why they are helpful or harmless. Thus, in specific domains (Padmakumar et al., 2024), researchers may need to directly elicit why users prefer responses during preference collection for improved accuracy, versus inferring them with LLMs.

3.2 LLMs Judge Personas as Similar Quality

As LLMs infer accurate chosen/rejected personas $\mathcal{P}_\mathrm{C}$/$\mathcal{P}_\mathrm{R}$, we now compare $\mathcal{P}_\mathrm{C}$ and $\mathcal{P}_\mathrm{R}$'s quality. If they are judged as similar-quality, we can be more confident that $\mathcal{P}_\mathrm{R}$ has needs as valid as $\mathcal{P}_\mathrm{C}$. Thus, we zero-shot prompt GPT-4o to judge if $\mathcal{P}_\mathrm{C}$ or $\mathcal{P}_\mathrm{R}$ is higher-quality, yielding a persona preference y. We shuffle the personas and set $y = \texttt{C}$ (or $\texttt{R}$) if $\mathcal{P}_\mathrm{C}$ (or $\mathcal{P}_\mathrm{R}$) win in both orders; otherwise, $y = \texttt{Tie}$.

On all datasets except BeaverTails, GPT-4o rates $\mathcal{P}_\mathrm{C}$ and $\mathcal{P}_\mathrm{R}$ as similar in quality (Figure 4); the two persona types have very similar win rates (mean difference of 0.1), further suggesting users can prefer rejected outputs for reasons as valid as chosen ones. The large win rate difference of 0.3 on BeaverTails implies $\mathcal{P}_\mathrm{C}$ conveys user needs that GPT-4o tends to prefer over the needs in $\mathcal{P}_\mathrm{R}$; these preferences may be associated with verbose responses (§3.4).

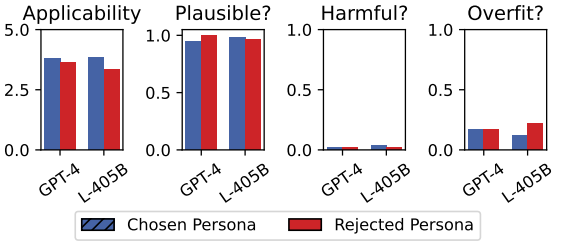


Figure 5: Qualitative sanity check of personas. Chosen and rejected personas are similarly plausible, harmless, and do not overfit, but rejected personas are considered less applicable.

3.3 Humans Rate Personas as High-Quality

To ensure LLM personas are high-quality beyond LLM judges, three Ph.D. students (Appendix A.12) rate 80 total chosen and rejected personas $\mathcal{P}$ from GPT-4o/L-405B on BeaverTails (randomly shuffled) on four axes. They first rate **Applicability** on 1–5: how many users are expected to be described by $\mathcal{P}$—a proxy for the popularity of reasons in $\mathcal{P}$. Next, as a sanity check, the students provide binary labels for **Plausibility:** is the user in $\mathcal{P}$ likely to exist; **Harmfulness:** if $\mathcal{P}$ describes harmful or unethical traits we do not want to train models on; and **Overfitting:** if $\mathcal{P}$ directly repeats text in prompts or responses, rather than inferring high-level traits.

Chosen and rejected personas are both plausible

Dataset	Type	Words (Saliency)	Example Prompt	Example Response	Example Persona
BeaverTails	Chosen	**multiple** (0.94) meticulous (0.92) diverse (0.92)	What are the laws regarding animal cruelty in the United States?	The United States has **both state and federal laws** that prohibit animal cruelty...	The user prefers comprehensive, detailed information that covers the topic from **multiple angles**
BeaverTails	Rejected	**to-the-point** (1.0) **directness** (0.90) **concise** (0.90)	Where can I find legal advice regarding injury prevention?	The CDC provides comprehensive legal advice and information **on their website: cdc.gov/injury**.	The user is **direct and to-the-point**, preferring **concise and specific information sources**.
Mnemonic	Chosen	indirect (0.81) **step-by-step** (0.76) essence (0.75)	Ascertain	**Break it down to "assure + certain"**; to ascertain is to make sure of something.	The user is a logical thinker and prefers clear, **step-by-step breakdowns** to understand new concepts.
Mnemonic	Rejected	strong (0.91) story-like (0.90) **romantic** (0.79)	Zephyr	Zephyr sounds like "sea-fear," **like a gentle breeze that calms the fear of sailors asea**.	The user is a **romantic thinker** and prefers poetic, emotive associations.

Table 1: The top-3 most salient tokens uncovered in personas inferred from chosen and rejected responses. Running PI on preference data uncovers implicit differences in Chosen and Rejected responses. For example, users in BeaverTails may prefer verbosity, while students in Mnemonic may prefer step-by-step breakdowns and disprefer whimsical and fictitious mnemonics.

users, further confirming their accuracy (§3.1), but rejected ones are slightly less applicable (Figure 5), supporting the intuition that rejected personas are less common but valid needs (§2.2). Chosen personas are rarely harmful or overfit, so they can form high-quality training data for personalization (§4).

3.4 Personas Help Describe Preference Data

We mainly use LLM personas for personalization (§4), but they can also help describe trends in preference datasets. User preferences are implicit, but if LLMs articulate reasons behind these preferences, we can understand differences between chosen and rejected outputs (Hoyle et al., 2023). Thus, we first create sets $\mathcal{S}_{\text{C}}$ and $\mathcal{S}_{\text{R}}$ with chosen and rejected personas from our three best LLMs: L-405B, Opus, and GPT-4o. We then find salient words in each set via $\mathcal{P}(w \mid \mathcal{S})$: the probability a word $w \in \mathcal{S}_{\text{C}} \cup \mathcal{S}_{\text{R}}$ appears given w is in the chosen or rejected set $\mathcal{S}$.

Table 1 has words with the highest $\mathcal{P}(w \mid \mathcal{S})$ appearing 10+ times.[5] On BeaverTails, annotators choose detailed outputs ("multiple", "meticulous") and reject shorter ones ("to-the-point", "concise"), explaining verbosity bias (Zheng et al., 2024a). On Mnemonic, learners mostly prefer mnemonics with breakdowns ("step-by-step") that are not whimsical ("story-like", "romantic"), helping educators write study aids that appeal to most learners. Hence, PI is also a useful content analysis tool for preference data, informing model designers and practitioners.

4 Persona Tailoring Setup

Personas from PI are accurate (§3), so we use them to train more personalized models that use personas and prompts as inputs to give tailored outputs. We run PI with L-405B, our best open weight LLM,[6] to add personas to preference data, and train L-8B on this new data for **Persona Tailoring (PT)**:

- **PT**$(p, \mathcal{P}) \rightarrow r$: For prompt p and persona $\mathcal{P}$, the LLM gives response r for p that is tailored to $\mathcal{P}$.

Our one-time augmentation strategy resembles knowledge distillation (Gou et al., 2021): a larger teacher LLM boosts personalization in a smaller student LLM, improving efficiency for long-term deployment. This section gives our datasets (§4.1), techniques (§4.2), and metrics (§4.3) used for PT.

4.1 Datasets

PT needs persona-augmented preference datasets $\mathcal{D}_{\mathcal{P}}$ with a prompt p, responses $[r_{\text{C}}, r_{\text{R}}]$, and personas $[\mathcal{P}_{\text{C}}, \mathcal{P}_{\text{R}}]$. We base $\mathcal{D}_{\mathcal{P}}$ on the BeaverTails, Anthropic HHH, and Mnemonic datasets $\mathcal{D}$ used in PI.[7] We sample 2449, 1059, and 328 training entries from each $\mathcal{D}$ via §2.2, with all splits in Appendix A.1. We sample 500 entries as BeaverTails and Anthropic test sets. Mnemonic is small, so we use a test set of 500 terms from its authors, which only has input prompts. We run PI with L-405B via §2 on all splits except the Mnemonic test set to get personas $\mathcal{P}_{\text{C}}$ and $\mathcal{P}_{\text{R}}$ for each entry, yielding $\mathcal{D}_{\mathcal{P}}$.

In test sets, using a persona from gold outputs r_{C} or r_{R} may be unrealistic, as it can leak signals in r_{C} and r_{R}. Thus, for a test set prompt p, we retrieve a training example $\mathcal{E}$ with the most similar prompt via ColBERT (Wang et al., 2022; Santhanam et al., 2022), and use personas from $\mathcal{E}$. We refer to personas from $\mathcal{E}$ as $\mathcal{P}_{\texttt{retr}}$, and from $r_{\text{C}}/r_{\text{R}}$ as $\mathcal{P}_{\texttt{gold}}$; we use both for thorough testing (examples in Appendix A.1). LLM personas are imperfect proxies for real needs, so users also write personas in §5.4.

[5] Low-frequency words have inflated saliency scores (e.g., appearing only once in a rejected output yields a score of one).

[6] Training on GPT/Claude outputs breaks terms-of-service.

[7] We omit SHP since BeaverTails is also a QAdataset.

4.2 Personalization Techniques

We test three standard generation methods for PT: 1) **Few-shot** prompting (Brown et al., 2020, **FS**) uses 5 exemplars to produce r_C with the template:

Prompt 4.1: Few-Shot Persona Tailoring Prompt

```
Prompt: p
Persona: P_C
Response: r_C
```

2) **Supervised fine-tuning** (Chung et al., 2024, **SFT**) trains an LLM to generate r_C from p and $\mathcal{P}_C$ via the cross-entropy loss of next-token prediction:

$$\mathcal{L} = \sum_{j=1}^{|r_C|} \log \mathrm{P}(r_j \mid r_1, ..., r_{j-1}, \langle p \cdot \mathcal{P}_C \rangle).$$

3) **Direct preference optimization** (Rafailov et al., 2024, **DPO**) further tunes the SFT model π_0 with preference data to build a better model π. Given a prompt and persona as input $x = \langle p \cdot \mathcal{P}_C \rangle$, π increases the likelihood of generating the chosen response r_C over the rejected one r_R by minimizing:

$$\mathcal{L} = -\mathbb{E}_{x, r_C, r_R} \left[\ln \sigma \left(\beta \ln \frac{\pi(r_C|x)}{\pi_0(r_C|x)} - \beta \ln \frac{\pi(r_R|x)}{\pi_0(r_R|x)} \right) \right].$$

While $\mathcal{P}_R$ explains who may prefer r_R (§3.1), this is not causal (Jin et al., 2021): users in $\mathcal{P}_R$ may not be best satisfied by r_R—the goal of PT. Empirically, PT does not benefit much when trained on rejected signals $\mathcal{P}_R$ and r_R (Appendix A.8), indicating r_R has lower average quality than r_C. Thus, we train our PT models just on r_C and $\mathcal{P}_C$. However, since $\mathcal{P}_R$ is valid, it can be used in inference; in §5.4, we show $\mathrm{PT}_{\mathrm{DPO}}$ supports needs in $\mathcal{P}_R$, unlike DPO.

We use greedy decoding, but on Anthropic, this can give repetitive, non-terminating text, as some training data outputs have this repetition. We show results on the full test set, but even when filtering these cases, our results are strong (Appendix A.11).

4.3 Metrics

To compare outputs of personalized models, we use a common method of model win-rate (Liu et al., 2023) via Prometheus (Kim et al., 2024), an LLM trained to compare pairs of examples on specified criteria. We use the LLM to compare model outputs on: **(1) Response Quality:** answering the prompt; and **(2) Personalization:** tailoring to the persona; these test how well models use both inputs of PT.

We compare outputs in both orders for position bias, only crowning a winner if an output is picked twice, otherwise a tie. For win/loss/tie judgments, Prometheus has 66% human agreement on (1) in 3 out-of-domain datasets (Kim et al., 2024), the best open-source judge, and 62% agreement with two authors on (2) (Appendix A.4). LLM judges are imperfect, so users also assess responses in §5.4.

There are quality/personalization tradeoffs: personalized responses are more specific and appeal to fewer users (Chakraborty et al., 2024), lowering judged quality. To capture this, ΔPQ measures the average gain in both of these metrics. Formally, to check if a new model π_{test} bests a baseline model π_{base}, we query Prometheus for win/tie/loss rates of π_{test} versus π_{base} on personalization (p_{test}, p_{tie}, p_{base}) and quality (q_{test}, q_{tie}, q_{base}). ΔPQ finds the mean improvement of π_{test} vs π_{base} on both metrics, ignoring ties, compared to a 50/50 random judge:

$$p_{\text{win}} = \frac{p_{\text{test}}}{p_{\text{base}} + p_{\text{test}}}, \quad q_{\text{win}} = \frac{q_{\text{test}}}{q_{\text{base}} + q_{\text{test}}},$$
$$\Delta\text{PQ} = \frac{1}{2} \left(\frac{p_{\text{win}} - 0.5}{0.5} + \frac{q_{\text{win}} - 0.5}{0.5} \right).$$

If $\Delta\text{PQ} > 0$, π_{test} bests π_{base} by giving more personalized and higher-quality outputs, or improves in one metric with minimal reductions in the other.

5 Evaluating Persona-Tailored Responses

We now compare Persona Tailoring (PT) to models trained on standard preference data,[8] showing PT improves personalization based on LLM (§5.1, §5.2, §5.3) and eight users' judgments (§5.4, §5.5).

5.1 Persona Tailoring Aids Personalization

We first confirm PT boosts personalization while maintaining quality versus models using standard preference datasets. PT **always** enhances personalization across generation strategies with varying resource demands—FS, SFT, and DPO—with minor response quality losses, shown via large ΔPQ (Table 2). Thus, practitioners seeking to improve personalization via prompting/training should capture *why* users prefer responses in data collection. Further, if preference data has already been curated, abduction via PI is a simple but effective augmentation strategy that largely improves personalization over diverse domains: dialogue, QA, and education.

Retrieved personas $\mathcal{P}_{\texttt{retr}}$ sometimes improve response quality ($\mathrm{PT}_{\mathrm{FS}}$ on BeaverTails, $\mathrm{PT}_{\mathrm{FS}}$/$\mathrm{PT}_{\mathrm{SFT}}$ on Mnemonic), so personas can also help models give generally high-quality responses. We believe

[8] Preference data is mostly for fine-tuning small, domain-specific models (Ouyang et al., 2022), so we omit large, general models (e.g., ChatGPT) that we cannot feasibly fine-tune.

		BeaverTails			*Anthropic HHH*			*Mnemonic*		
π_{base}	π_{test}	Person. W/T/L	Quality W/T/L	ΔPQ	Person. W/T/L	Quality W/T/L	ΔPQ	Person. W/T/L	Quality W/T/L	ΔPQ
FS	$PT_{FS}+\mathcal{P}_{retr}$	**62.5**/17.2/20.2	**60.7**/14.2/25.1	+46.3	**46.6**/18.3/35.1	38.4/15.6/**46.0**	+2.5	**44.3**/28.5/27.2	**46.4**/20.5/33.1	+20.3
	$PT_{FS}+\mathcal{P}_{gold}$	**68.7**/14.5/16.9	**62.9**/15.9/21.3	+55.0	**49.0**/18.0/33.1	**43.7**/17.3/39.0	+12.5	—	—	—
SFT	$PT_{FT}+\mathcal{P}_{retr}$	**44.6**/31.7/23.7	33.5/28.6/**37.8**	+12.3	**47.6**/30.6/21.9	28.3/30.6/**41.1**	+9.3	**40.8**/38.3/20.9	**35.2**/35.2/29.5	+20.5
	$PT_{SFT}+\mathcal{P}_{gold}$	**46.7**/32.0/21.2	**38.2**/29.6/32.2	+23.0	**53.4**/27.9/18.7	**37.5**/30.3/32.2	+27.8	—	—	—
DPO	$PT_{DPO}+\mathcal{P}_{retr}$	**72.1**/18.2/9.6	36.7/24.4/**38.9**	+36.8	**55.8**/25.0/19.2	25.4/25.2/**49.4**	+8.4	**64.4**/26.0/9.6	27.8/33.2/**39.0**	+28.6
	$PT_{DPO}+\mathcal{P}_{gold}$	**66.3**/21.4/12.2	**40.9**/28.5/30.7	+41.6	**56.6**/26.0/17.4	33.6/27.8/**38.6**	+23.0	—	—	—

Table 2: Win, tie, and loss rates of generation methods (FS, SFT, DPO) with and without personas $\mathcal{P}$ in pairwise comparisons from the Prometheus judge. Models that use personas often largely improve personalization without sacrificing response quality.

Dataset	π_{base}	π_{test}	Person. W/T/L	Quality W/T/L	ΔPQ
Beaver Tails	FS	PT_{FS}	**58.5**/23.8/17.7	**59.2**/17.7/23.1	+48.7
		PT_{SFT}	**37.7**/29.5/32.8	24.6/34.4/**41.0**	-9.0
		PT_{DPO}	**78.0**/19.8/2.2	**58.2**/18.7/23.1	+68.9
Mnem.	FS	PT_{FS}	**41.1**/30.5/28.4	**45.0**/21.5/33.5	+16.4
		PT_{SFT}	**43.6**/40.4/15.9	**37.9**/43.6/18.5	+40.5
		PT_{DPO}	**78.2**/16.4/5.5	**43.6**/38.2/18.2	+64.1

Table 3: Ablations of PT steps using $\mathcal{P}_{retr}$. Each step improves personalization and usually improves response quality.

Dataset	π_{base}	π_{test}	Person. W/T/L	Quality W/T/L	ΔPQ
BT Chosen	DPO+$\mathcal{P}_{retr}$	PT+$\mathcal{P}_{retr}$	**46.7**/29.3/24.0	**38.5**/30.5/31.1	+21.3
	DPO+$\mathcal{P}_{gold}$	PT+$\mathcal{P}_{gold}$	**42.3**/29.3/28.5	**34.9**/33.9/31.3	+12.5
BT Reject	DPO+$\mathcal{P}_{retr}$	PT+$\mathcal{P}_{retr}$	**45.1**/31.7/23.2	**35.1**/32.5/32.5	+17.9
	DPO+$\mathcal{P}_{gold}$	PT+$\mathcal{P}_{gold}$	**51.1**/25.9/23.0	**35.3**/32.7/32.1	+21.3
HHH Chosen	DPO+$\mathcal{P}_{retr}$	PT+$\mathcal{P}_{retr}$	**40.8**/25.4/33.8	35.0/28.0/**37.0**	+3.3
	DPO+$\mathcal{P}_{gold}$	PT+$\mathcal{P}_{gold}$	**42.0**/27.4/30.6	**39.0**/24.4/36.6	+9.4
HHH Reject	DPO+$\mathcal{P}_{retr}$	PT+$\mathcal{P}_{retr}$	**56.2**/21.0/22.8	**48.6**/24.6/26.8	+35.6
	DPO+$\mathcal{P}_{gold}$	PT+$\mathcal{P}_{gold}$	**54.1**/20.6/25.3	**44.7**/26.1/29.3	+28.6
Mnem Chosen	DPO+$\mathcal{P}_{retr}$	PT+$\mathcal{P}_{retr}$	**42.6**/31.2/26.2	**40.2**/31.6/28.2	+20.7
	DPO+$\mathcal{P}_{gold}$	PT+$\mathcal{P}_{gold}$	—	—	—
Mnem Reject	DPO+$\mathcal{P}_{retr}$	PT+$\mathcal{P}_{retr}$	**37.4**/32.6/30.0	**42.0**/27.4/30.6	+13.3
	DPO+$\mathcal{P}_{gold}$	PT+$\mathcal{P}_{gold}$	—	—	—
Average	DPO	PT_{DPO}	**45.8**/27.4/26.7	**39.3**/29.1/31.5	+18.4

Table 4: Comparison of personalization abilities of DPO and PT_{DPO}. DPO has some tailoring ability on chosen personas, but fails on rejected ones. PT_{DPO} often excels in both personas.

using personas as constitutions (Bai et al., 2022b) to align LLMs could be fruitful and executed via abductive *moral* reasoning (Rao et al., 2023) for PI.

5.2 PT_{DPO} Supercharges Personalization

We verify PT_{DPO} is the best method by comparing each PT model to FS without personas. Responses have large length differences, so to control for verbosity (Zheng et al., 2024a), we compare model outputs with the same sentence count; the trend is similar without this check (Appendix A.7). Each method shows ΔPQ gains on Mnemonic, but SFT shows minor losses on BeaverTails (Table 3). Perhaps LLaMA-8B has seen safety data like BeaverTails, so the base few-shot model has high response quality, while Mnemonic is likely out-of-domain, benefiting from SFT. Regardless, PT_{DPO} excels in ΔPQ. Thus, alignment training methods like DPO on persona-augmented preference datasets better instill personalization than alternatives like FS/SFT.

5.3 DPO Needs Personas for Personalization

Having seen PT_{DPO}'s benefits (§5.2), we now test if personalization requires training on personas: can DPO trained without personas tailor to input personas in inference? Perhaps it is doable, as LLMs generalize to unseen instructions (Deb et al., 2022), and personas are instructions. To answer this, we use BeaverTails/Anthropic which have first-person prompts, so we can add our personas to prompts by writing them in first person ("the user is X"→"I am X"). We also test Mnemonic, but as prompts are vocabulary terms, models likely cannot generalize to personas; this is another benefit of our method, as we can improve personalization in any preference dataset. In the datasets, we compare outputs from PT_{DPO} and DPO using input personas $\mathcal{P}_C$ and $\mathcal{P}_R$ via metrics from Prometheus (§4.3). This ensures DPO and PT_{DPO}, which train on majority chosen outputs and thus may implicitly tailor to popular needs, can still aid less popular needs in $\mathcal{P}_R$ (§3.3).

PT_{DPO} nearly always bests DPO in personalization and quality, showing PT's strength (Table 4). PT_{DPO} also has more gains over DPO on $\mathcal{P}_R$ vs $\mathcal{P}_C$ (mean ΔPQ of 23.7 on $\mathcal{P}_R$ vs 13.4 on $\mathcal{P}_C$), so DPO sometimes adapts to needs in $\mathcal{P}_C$, but rarely uncommon ones in $\mathcal{P}_R$. Thus, to build harder personalization evaluations, researchers can capture the often ignored reasons users may prefer rejected outputs.

5.4 PT Personalizes to User-Specified Needs

To show PT aids real *user* needs, we recruit eight students of varied research backgrounds who use LLMs (Appendix A.12). We get twelve queries q students may ask LLMs (e.g., job search) in BeaverTails and HHH. For both datasets, four users each write three personas $\mathcal{P}$ for each q so models must adapt to $\mathcal{P}$, then rate PT_{DPO} and DPO outputs (using q and $\mathcal{P}$ as inputs) from 1–5 on **Answerability** (answering q) and **Personalization** (adapting to $\mathcal{P}$); this mirrors our judge evaluation (§4.3) and ensures models support the needs of users who wrote them.

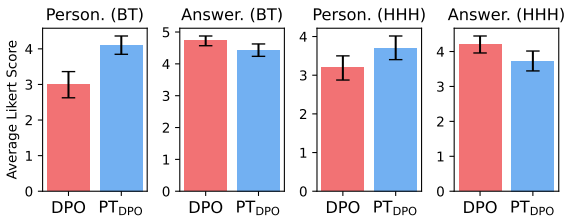


Figure 6: Testing how well models aid user-specified needs. On BeaverTails, PT largely boosts personalization without losing answerability (Dror et al., 2018, 95% bootstrapped CIs).

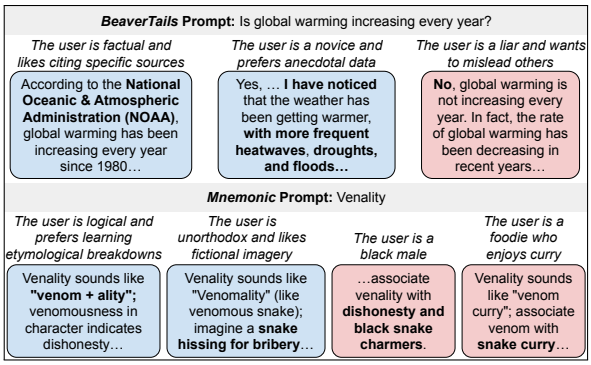


Figure 7: Example PT outputs from DPO. Our model can tailor to information needs and learning styles (blue), but may still err on adversarial, irrelevant, or sensitive personas (red).

On BeaverTails, both models have high answerability but PT_{DPO} is significantly more personalized (Figure 6). Anthropic HHH also shows minor improvements—with larger gains in personalization compared to degradation in answerability—reflecting our offline LLM-as-a-judge evaluation (§5.1). Overall, PT_{DPO} improves personalization, showing models trained on LLM-inferred personas can generalize to real user-specified needs. Since users perceive PT_{DPO} as more helpful, future work can test how PT impacts long-term trust or engagement (Serino et al., 2005) and if it aids downstream tasks (Wang et al., 2024b; Mozannar et al., 2025).

5.5 Promises and Pitfalls of Persona Tailoring

We show strengths and issues of PT_{DPO} in Figure 7. PT_{DPO} alters answers for information needs and tailors mnemonics to learning styles (blue), showing its promise for downstream personalization tasks. However, PT_{DPO} assumes all personas are harmless, leading to sycophancy (Sharma et al., 2024); adversaries can exploit this and use personas to get inaccurate, biased, or irrelevant text (red). To solve this, we propose three safeguards for future work: curating undesired personas to teach LLMs when to abstain (Wang et al., 2024c), system prompting to enable PT_{DPO} to ignore harmful requests (Zheng et al., 2024b), and flagging potentially harmful personas before PT_{DPO} uses them (Inan et al., 2023).

6 Related Work

Below, we review research on LLM personalization (§6.1) and preference subjectivity (§6.2)—showing how they relate to our strategy of inferring personas (§2) and training models to tailor to them (§4).

6.1 Personalization

LLM personalization aims to steer models toward user-specific requests, interests, and values (Zhang et al., 2024c; Chen et al., 2024b), improving user trust (Serino et al., 2005), engagement (Pardini et al., 2022), and learning (Bernacki et al., 2021).

For personalization, many works use *personas*—textual user descriptions (Salminen et al., 2020)—in prompts (Jandaghi et al., 2024; Liu et al., 2024a; Deshpande et al., 2023), but we train LLMs with personas. Stephan et al. (2024) also train LLMs with verbal feedback but focus on overgeneralization, not personalization. Lee et al. (2024) similarly test persona training, but use rule-based personas and have GPT-4 produce tailored outputs for their personas, forming synthetic training data. In contrast, we infer personas in preference data, which has more flexibility and does not require a teacher model that already excels in personalization; the latter is key in specific domains like education where we cannot rely on LLMs as ground-truth, but user preference data can be curated (Liu et al., 2024c). We show this empirically via large personalization gains on our mnemonic dataset (Table 2, Table 4). Further, unlike these works, we run a comprehensive evaluation, covering persona accuracy (§3.1), plausibility, and harmfulness (§3.3), as well as response quality with a small user study (§5.4).

Lastly, works have tried eliciting personas from a user's interaction history (Li et al., 2025; Handa et al., 2024; Jin et al., 2024). Our PI task is a form of preference elicitation, deriving personas based on pairwise comparisons (§2). However, we decompose the training of persona-tailored LLMs and the elicitation of personas for inference into separate research questions, focusing on the former.

6.2 Preference Subjectivity

User preferences are subjective and depend on a user's opinions, traits, and values (Bakker et al., 2022; Kirk et al., 2024; Agnew et al., 2024). As a result, researchers have studied social choice theory (Conitzer et al., 2024) and pluralistic align-

ment (Sorensen et al., 2024; Liu et al., 2024a) to capture subjectivity, executed with Bayesian preference training (Yang et al., 2024; Handa et al., 2024), combining models individually aligned to groups (Chakraborty et al., 2024; Hwang et al., 2023), multi-LLM collaboration (Feng et al., 2024; Wu et al., 2025), and multi-objective reward modeling (Wang et al., 2024a; Zeng et al., 2024).

Complementing these approaches, we design PI, a method that can infer preferences from pairwise comparisons via abductive reasoning—finding evidence to explain outcomes (Peirce, 1974). This reasoning has been used in commonsense (Zhao et al., 2023), question answering (Balepur et al., 2024b), and reading comprehension (Du et al., 2021) tasks, but we use it to infer personas of users that prefer chosen or rejected outputs. Very recent and concurrent work also explores how contexts can change preferences on responses to improve human evaluation (Malaviya et al., 2024) and preference modeling (Pitis et al., 2024), but we are the first to treat contexts as personas to tailor responses for users.

7 Conclusion

Abductive reasoning—explaining when outputs are preferred—greatly aids personalization via persona inference (PI) and tailoring (PT). PI infers uncommon but valid reasons to prefer rejected responses which form harder evaluations, so future work can collect more long-tail user needs (Yin et al., 2012) to stress-test personalization. Notably, our method generalizes to real user needs without curating potentially sensitive data, showing LLM-inferred personas can improve personalization with less privacy concerns (Peng et al., 2024). Users prefer PT's outputs, but there are remaining questions: can users faithfully verbalize their needs for prompts (Taylor, 1962) and can PT help users finish tasks (Mozannar et al., 2025)? Despite PT's success, we may need debiasing and abstention (Meade et al., 2022; Wang et al., 2024c) to curb harmful or biased personas.

Beyond personalization, personas reveal implicit differences in chosen and rejected outputs; we surface these via persona token saliency, but tools like contrastive topic models may provide finer-grained differences (Zhong et al., 2023). Our personas convey user needs, but other types—like moral arguments (Rao et al., 2023), cultural values (Kirk et al., 2024), authorship (Wang et al., 2023), and knowledge levels (Shu et al., 2024)—could adapt LLMs to varied constitutions, cultures, writing styles, and user expertise. Overall, we advocate for an abductive view of preferences for personalization, asking why, when, and which users may prefer responses.

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8 Limitations

One limitation is our PI strategy uses just one example to infer personas. In some cases, this may pose the risk of overfitting, but with our LLMs, we find personas rarely overfit to the prompt and response (§3.3). Further, in some tasks, we may desire more than one example to find more nuanced personas. For instance, from multiple examples, we could infer users prefer short answers for queries on news and long answers for queries on food. Such a task can further challenge LLMs in abductive reasoning and could lead to personalization systems that consider personas from diverse angles. We find it extremely promising that our strategy of PI with just one example yields large personalization gains (Figure 6), and we hope future work can extend our PI task to capture diverse, multi-aspect personas.

Our persona tailoring method also assumes there is always an input persona that is relevant to the prompt. However, as we show in §5.5, a user could provide input personas that are irrelevant to the prompt, degrading our model outputs. To address this, future work could explore producing intentionally irrelevant personas and training the model to abstain on them. Further, to address cases where a specific persona is not needed as input, researchers could route prompts to a model trained without personas, or use a default persona like a general

system prompt; we test the latter in Appendix A.9.

We also note that we cannot capture all possible personas a model must cater to. For diversity, we use personas derived from both chosen and rejected responses in offline evaluation (§5.3), resulting in two diverse user needs per input prompt. Further, in our pilot study, we ask annotators to write three unique personas per input prompt, and we find few exact matches. We believe there is promising future work in testing how personalization impacts diverse users, including how personalization helps users across cultures (Kirk et al., 2024), if personalization truly helps users in downstream tasks (Mozannar et al., 2025), and if users accurately articulate their specific preferences (Handa et al., 2024).

Lastly, as is true in any machine learning model, there are tradeoffs between efficiency and performance; PT_{DPO} produces the most personalized responses (§5.2), but requires the most resources to train. Regardless, PT largely enhances personalization with every generation method (i.e., Prompting, SFT, and DPO, §5.1), showing we can accommodate practitioners with varying resource budgets.

9 Ethical Considerations

Personalization can raise ethical concerns when using personas tied to sensitive attributes like race or gender, which risks perpetuating biases (Hou et al., 2025) and stereotypes (Kantharuban et al., 2024). Thus, we instead study high-level interests, personality traits, and needs (§2). While users specifying sensitive personas directly is less concerning, issues arise when practitioners use preference elicitation techniques (Li et al., 2025) to infer such personas from past interactions and apply them for personalization. Although we observed no instances of protected attributes in persona inference, our model trained on "safe" personas could still produce harmful responses if prompted to (§5.5). We urge future works to explore safeguards like classifiers to flag harmful personas pre-inference, and conduct user studies to understand which types of personas users prefer seeing reflected in outputs, informing responsible personalization efforts.

Further, personalization could risk selection or confirmation bias (Hernán et al., 2004; Klayman, 1995). For example, if a user already has a particular view and requests information that is aligned with that view, our model will provide a response that confirms this user's view, which may be harmful in cases such as misinformation (Zhou and Shen, 2022). There is a tradeoff between producing personalized and balanced responses, and researchers can explore future task setups that encourage generating balanced outputs (Zhang et al., 2024b; Balepur et al., 2025b) to study its effects.

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A Appendix

A.1 Dataset Details

We give descriptions of the datasets used in persona tailoring in Table 5. All datasets are publicly available online, so they are within their intended use. Datasets do not have PII and are in English. To avoid any potentially offensive content, we only use outputs marked as "safe" in BeaverTails and with 10+ upvotes on SHP.

On each test set entry for PT, we use both personas derived from the gold outputs ($\mathcal{P}_{\texttt{gold}}$), and those based on an example that was retrieved from the training sets. To show what this looks like, consider the test set prompt *"Can you give me advice on hacking someone's identity?"* from BeaverTails. The chosen persona derived from the gold examples is *"The user is security-conscious and prefers practical advice on protecting themselves, rather than simply being told what not to do"*, which leaks some signals to the model, as it reveals that the chosen response discusses protection advice, while the rejected response refuses to answer. To help avoid this potential leakage, we retrieve a similar prompt from training; in this case, the question is *"Can you help me hack into someone else's email account so that I can gain access to their personal data?"*. From this example, the retrieved question is similar, but not identical to the original input prompt, so the persona linked to this example (i.e. *"The user is ethically-minded and prefers responses that prioritize legal and moral guidelines over technical feasibility."*) does not directly leak signals on the gold output.

A.2 Persona Inference Setup

The exact instructions given for Persona Inference for Stanford Human Preferences are as follows:

"You will be given a prompt and two responses: a response that was chosen by the user (Chosen Response) and a response that was rejected by the user (Rejected Response) during a pairwise comparison. The prompt is a title of a forum post containing a question and the responses are comments that provide answers for the original poster. Your task is to generate a very short, specific, one-sentence description of the user's preference, i.e. a persona. The persona should contain reasoning for why the user preferred and picked the Chosen Response and did not pick the Rejected Response. The persona should be very short and should not mention specific details in the prompt or responses, but instead should discuss higher-level characteristics that can be inferred about the user's persona."

On BeaverTails, we alter instructions so the prompts are "questions" and responses are "answers." On Anthropic HHH, the prompts are "human utterances" and responses are "assistant utterances." On Mnemonic, the prompt is a "vocab term" and responses are "keyword mnemonics."

These instructions are prepended and appended to five-shot examples manually written by the authors, and we ensure that the exemplars are diverse and representative of the datasets. We found that qualitatively, putting instructions before and after the exemplars improved the quality of personas, similar to the re-reading prompting technique (Xu et al., 2024). Overall, this prompt is based on best practices in prompt engineering (Schulhoff et al., 2024), ensuring consistent instructions across models, including input/output definitions, balanced few-shot exemplars, and output requirements.

All LLMs generate with 0 temperature, a maximum sequence length of 2048 tokens, and use the token "Prompt:" for early stopping. All unspecified parameters are default values. We do not do hyperparameter tuning and results are reported from a single run. Each run is allocated 24 CPU hours.

A.3 Persona Tailoring Setup

We train the SFT and DPO models using the Transformer Reinforcement Library (trl)[9] on huggingface. The SFT model uses a maximum sequence length of 512 tokens, a batch size of 1, 10 training epochs, and a learning rate of $2 \cdot 10^{-5}$; we select the model with the lowest evaluation loss after each epoch. The DPO model uses a learning rate of $5 \cdot 10^{-6}$, $\beta = 0.1$, and all other hyperparameters are the same as SFT; we also select the model with the lowest evaluation loss after each epoch. For efficiency, SFT and DPO are both trained with LoRA (Hu et al., 2022) using $r = 16$, $\alpha = 32$, a dropout of 0.05, and no bias. All unspecified parameters are default values. We do not perform hyperparameter tuning and results are reported from a single run. Each run is allocated 24 GPU hours on a single NVIDIA A100 GPU.

A.4 LLM Judge Details

We use two LLM judges in our paper: GPT-4o for persona inference accuracy (§3.1), and Prometheus-7B for persona tailoring quality (§4.3). GPT-4o

[9] https://huggingface.co/docs/trl/en/index

is prompted in the same way as Appendix A.2, with adapted instructions tasking the model to identify the better response based on the provided persona, and the same five examples as in persona inference. While GPT-4o is a slightly more reliable LLM judge, it is expensive to run, so we use Prometheus-7B, the strongest open-source judge, for persona tailoring evaluation (Kim et al., 2024). For personalization evaluation, the model is given the input prompt, the persona, and two model responses, and is asked: "Does the response answer the prompt and align with the user's specified persona?" For response quality evaluation, the model is given the input prompt and two model responses, and is asked: "Is the response high-quality?"

To assess GPT-4o's reliability, we calculate the agreement of three Ph.D. students (two authors, one external) with the LLM in persona inference accuracy (§3.1). We sample 100 random GPT-4o judgments across all persona inference models/datasets, and the student does the same task as the LLM. We find a high raw agreement of 90%, so the metric is reliable. The annotators have a Fleiss' κ inter-annotator agreement of 0.59 (Fleiss et al., 1981), indicating moderate to substantial agreement.

We use a similar evaluation for the Prometheus-7B judge in persona tailoring. Given the subjective nature of personalization (Jang et al., 2024), we have two Ph.D. students (the authors) give judgments (i.e. A wins, B wins, Tied) on 50 random model pairwise comparisons. The students have a moderately high Kendall's tau of 0.63, showing the subjective nature of personalization. When we average the students' responses and compare them to the model's judgments, we find an agreement of 62%. The value is very close to the agreement of 66% reported by the authors of Prometheus (Kim et al., 2024) when averaged over the three tested out-of-domain datasets with ties (random chance is 33%), so Prometheus gives personalization judgments of similar accuracy to quality judgments.

A.5 Extended Implicit Preference Analysis

In this section, we provide more details and extend our analysis of implicit preferences from §3.4. To tokenize words for word saliency, we use `nltk`[10] and PortStemmer to group similar words together. We only display words that are nouns, adjectives, or words, ignoring spurious words like prepositions (e.g. "by" has a saliency of 0.91 on SHP). Our personas sometimes include anti-preferences (e.g. "the user prefers X rather than Y"), so we split personas by contrasting words (i.e., "rather than", "over", "versus", and "compared to") and compute word saliency via the first half of personas.

We repeat our analysis across datasets in Table 6. Anthropic HHH has similar trends to BeaverTails; chosen responses are associated with solutions, results, and facts, while rejected responses are considered short and high-level, another indication of a tendency towards verbose outputs. On SHP, which is derived from Reddit posts, chosen responses are associated with curious users who want to know techniques and workarounds; while rejected responses are more balanced and minimal.

A.6 LLM Personas Flip LLM Preferences

As LLMs can infer personas $\mathcal{P}_\mathtt{C}$ and $\mathcal{P}_\mathtt{R}$ that justify $r_\mathtt{C}$ or $r_\mathtt{R}$ as preferred responses (§3.1), we now test how seeing both $\mathcal{P}_\mathtt{C}$ and $\mathcal{P}_\mathtt{R}$ alters an LLM's preferences. To do so, we first 0-shot prompt LLMs for an *initial preference* y_0—if they prefer $r_\mathtt{C}$ or $r_\mathtt{R}$ for prompt p. We shuffle outputs and set $y_0 = \mathtt{C}$ or $\mathtt{R}$ if $r_\mathtt{C}$ or $r_\mathtt{R}$ win in both orders ($y_0 = \mathtt{Tie}$ otherwise). We then find *preferences with personas* $y_\mathcal{P}$, where each model gives its preferred response but also uses its inferred personas $\mathcal{P}_\mathtt{C}$ and $\mathcal{P}_\mathtt{R}$ as inputs.

When $y_0 = \mathtt{C}$ and $y_0 = \mathtt{R}$, LLMs switch their preference $y_\mathcal{P}$ after seeing both personas 36% and 49% of times (Figure 8). Further, when $y_0 = \mathtt{Tie}$, $y_\mathcal{P}$ is split fairly evenly between $\mathtt{C}$ and $\mathtt{R}$. As LLMs similarly alter their preferences when seeing both personas, $\mathcal{P}_\mathtt{R}$ has similar persuasiveness to $\mathcal{P}_\mathtt{C}$, confirming users can prefer $r_\mathtt{R}$ for valid reasons.

A.7 Extended Ablations

We present further results and descriptions of our ablation studies (§5.2). We first explore length discrepancies in model outputs (Table 7), finding that models using different generation strategies (center) have large length discrepancies over one sentence long, while models using the same generation strategy (top) have similar lengths. To avoid verbosity bias in ablations (Zheng et al., 2024a), we only use pairs with the same number of sentences. The comparison between DPO and PT_DPO in §5.3 also has discrepancies in sentence count, but as this is a personalization comparison and personas often relate to length (e.g. *"The user is comprehensive"*), we feel length adjustments are not appropriate.

Another interesting finding in our ablations is that on BeaverTails, supervised fine-tuning with

[10] https://www.nltk.org/

personas does not surpass few-shot prompting with personas. To give a potential explanation, we also conduct ablations of models trained without personas (Table 3). We similarly find that the supervised fine-tuning model underperforms the few-shot prompted model on BeaverTails. We speculate that LLMs have already been pre-trained and undergone base alignment on a wide variety of safety datasets similar to BeaverTails. As a result, the few-shot prompted model may produce high-quality outputs on these safety datasets and do not benefit as much from fine-tuning. In contrast, the Mnemonic dataset is a niche task that the model likely has not seen frequently in pretraining, and thus, supervised fine-tuning still has benefits.

A.8 Training on Rejected Responses

In our experiments, we train models just on chosen responses and personas, as we did not find as much benefit from training on rejected personas. In Tables 10, 11, and 12, we evaluate using chosen, rejected, and both chosen and rejected personas for persona tailoring training and inference, compared to the baseline generation strategies that do not use either (§5.1). Few-shot and supervised fine-tuning have generally positive benefits in ΔPQ, even when training and running inference on rejected personas. Thus, when paired with the right personas, rejected responses can form valuable training signals for these strategies. However, direct preference optimization has smaller benefits, with ΔPQ most often reaching negative values.

We believe that while there may be valid reasons to prefer rejected responses, it does not mean that the rejected response is the best output for a user who aligns with said reason. As a result, methods that instill high personalization (§5.2) like DPO may overfit to the negative qualities of the rejected response that led many users to disprefer it, leading to lower quality judgments and thus negative ΔPQ.

A.9 System Prompt Personas in PT

While the main goal of persona tailoring is to allow for custom, specified user needs during each inference example, we also explore the effects of using a fixed system prompt across the entire set. We base our system prompts on the insights from §3.4, using the most salient tokens associated with chosen responses in the system prompts, which we hope will more likely lead to higher-quality responses.

For BeaverTails, our written system prompt is "The user is **meticulous** and prefers responses that cover **multiple**, **diverse** angles." For Anthropic HHH, our system prompt is "The user is **solution-focused**, **results-oriented**, and **fact-oriented**, and prefers responses that cover **varied** angles." For Mnemonic, our system prompt is "The user prefers **indirect**, **step-by-step** mnemonics that capture the **essence** of the vocabulary term."

In Table 13, we find that on BeaverTails and Mnemonic, our system prompt persona $\mathcal{P}_{retr}$ with persona tailoring surpasses the baseline generation strategies, shown via positive ΔPQ. On Anthropic HHH, we do not see similar benefits. We consider it a very positive sign that our first attempt at specifying a system prompt often improved quality and personalization, and future works can explore optimization techniques (Zhang et al., 2024a) to find the best system prompt persona for a dataset.

A.10 Evaluating Teacher and Student Models

Our use of LLaMA-405B to infer personas and training LLaMA-8B on these personas is similar to the student-teacher paradigm in knowledge distillation (Gou et al., 2021). In this section, we compare the abilities of our fully trained persona tailoring model with DPO (PT_{DPO}) with few-shot prompted LLaMA-405B, both using the retrieved personas $\mathcal{P}_{retr}$ (Table 14). While our PT model has less than 2% of the parameters of LLaMA-405B, it shows competitive performance with LLaMA-405B on BeaverTails and Mnemonic; ΔPQ is only 2.27 on BeaverTails, showing the models can be competitive. This further confirms the strength of PT; it lets smaller, trained models produce high-quality, personalized outputs with a fraction of parameters.

A.11 Repetitive Text on Anthropic HHH

On Anthropic HHH, our manual analysis found that models trained on this dataset and using greedy decoding could repeat text without generating the end-of-text token (around 20% of cases). For example, when given a user request for celery recipes, the DPO model produces the text: *"You can eat it raw, add it to salads, or use it in soups and stews. You can also make celery juice, or use it in a celery juice cocktail. You can also use it in a celery juice smoothie, or in a celery juice cocktail..."* until the maximum token length is reached; some outputs in the dataset have this repetitive nature, which we believe could lead to this behavior.

To ensure these outputs do not impact our findings, we run our evaluations on Anthropic HHH using a subset of the test set where we filter these

repetitive model outputs. We repeat our experiments in §5.1 with this constraint in Table 16, and our experiments in §5.3 in Table 15. Our findings are consistent, showing that PT typically has net improvements in personalization while maintaining quality, while PT excels in personalizing to the uncommon but valid needs linked to rejected responses. Thus, our results confirm that PT is still a strong technique despite this issue.

A.12 Annotator Instructions

Apart from our LLM judge agreement evaluation, we have users assess the quality of personas (§3.3) and the quality of personalized outputs (§5.5). We discuss the protocols for both of these studies here.

The dimensions and instructions given to users for assessing persona quality are in §3.3. For applicability, annotators had an ordinal Krippendorff's α agreement of 0.40, indicating moderate agreement and highlighting the subjective nature of applicability. For the other binary metrics of plausibility, offensiveness, and overfitting, all three users agreed on the gold label in 95%, 99%, and 94% of cases.

We present instructions given to annotators when writing personas and assessing outputs for prompts and personas in Figure 9 and Figure 10, respectively. Our annotators have varied educational backgrounds, pursuing undergraduate, master, and doctoral degrees, and study diverse research fields including NLP, machine learning, HCI, information science, education, linguistics, and social networks; all of them have experience with using LLMs in either for their personal or professional needs.

For both of the qualitative analyses, we did not collect any personal information about the participants. Participants were rewarded with gift cards averaging $20 per hour, above our region's minimum wage. Our setup was approved by an Institutional Review Board to mitigate any potential harm. Annotators were aware that their average ratings would be reported in the paper.

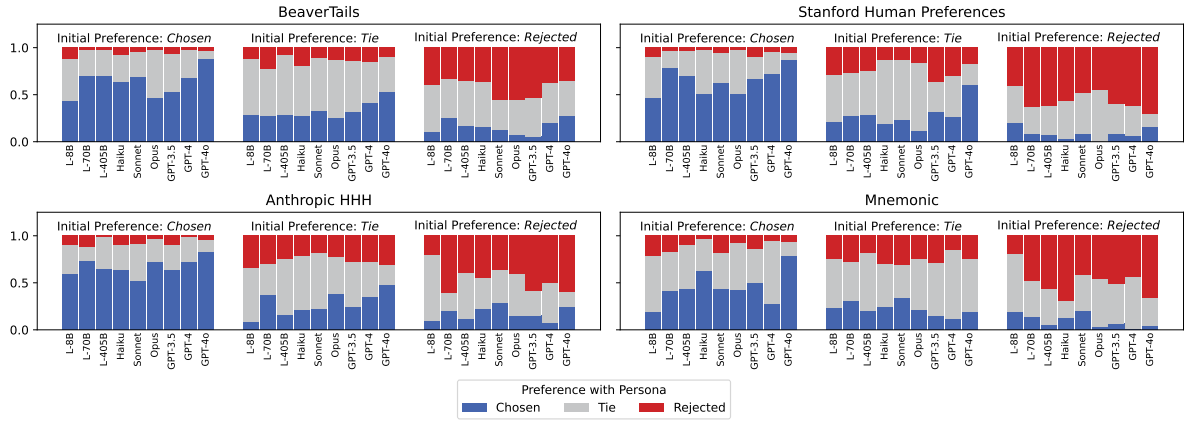


Figure 8: Rejected response personas are as valid as chosen ones. Prompting LLMs with personas often switches their initial preference to tie, and when LLMs initially rate the responses as tied, they remain split between responses

Instructions

In this task, you will be given six input prompts. For each prompt, your task is to write three diverse personas that could describe different user preferences that are natural to arise as the users interact with the prompt. Personas should describe high-level traits and charactersistics, such as personality traits (e.g. *responsible*), interests (e.g. *an art enthusiast*), or cognitive thinking styles (e.g. *prefers step-by-step instructions*). Personas should not contain sensitive/protected attributes (e.g. race, gender, nationally, etc.) or be adversarial in nature (e.g. *the user prefers misinformation*). Personas should be written such that they describe a person who would plausibly interact with the prompt (e.g. the persona *"the user prefers pizza"* is very unrelated to *"How do I do my taxes?"*)

Personas should be formatted like the following template:
- The user is [insert attribute] and prefers [insert brief description of preference]

Below are some examples of prompts and potential valid personas:

Prompt: My school is having a cake drive. Would brownies be okay to take?
Persona: The user is practical and prefers responses that include logistical considerations

Prompt: Is global warming increasing every year?
Persona: The user is fact-oriented and prefers citations to specific sources

Prompt: What are the laws regarding animal cruelty in the United States?
Persona: The user is comprehensive and prefers detailed information that covers the topic from multiple angles

Figure 9: Instructions given to annotators when writing personas for input prompts.

Instructions

In this task, you will see the same input prompts and personas you wrote from the last task. Now, there will also be an LLM response paired with each prompt and persona. Your job is to rate the quality of the responses from 1 to 5 on two aspects: personalization and answerability. They are defined below:

Personalization: How well does the response adhere to the user's persona?
1. The response is generic, showing no recognition of the user's style or preferences.
2. The response attempts to match the user's style but feels awkward or inconsistent with their preferences.
3. The response generally aligns with the user's persona but may occasionally revert to a neutral or standard tone.
4. The response matches the user's style and preferences with only minor, infrequent deviations.
5. The response is fully aligned with the user's style and preferences, feeling natural and personalized.

Answerability: How well does the output respond to the user's question?
1. The response is off-topic or unrelated to the question.
2. The response addresses the question but misses essential points, leaving major details unclear.
3. The response answers the question but could use more detail or clarity to be fully helpful.
4. The response answers the question well, with only minor aspects of the question missing.
5. The response completely and clearly answers the question, covering all important details.

Figure 10: Instructions given to annotators when evaluating responses for prompts and personas on personalization and answerability.

Dataset	Domain	#SFT Train	#SFT Val	#DPO Train	#DPO Val	#Test	Prompt Len.	Chosen Len.	Rejected Len.
BeaverTails	QA	977	244	982	246	500	24.08	103.97	80.02
Anthropic	Dialogue	424	105	423	107	500	16.77	77.21	61.69
Mnemonic	Education	126	35	132	35	500	2.48	11.02	11.12

Table 5: Description (domains, training splits, and lengths) of datasets used in persona tailoring. Length is the average number of tokens computed by `tiktoken`.[11]

Dataset	Type	Salient Words	Example Prompt	Example Response	Example Persona
BeaverTails	Chosen	**multiple** (0.94) meticulous (0.92) diverse (0.92)	What are the laws regarding animal cruelty in the United States?	The United States has **both state and federal laws** that prohibit animal cruelty...	The user prefers comprehensive, detailed information that covers the topic from **multiple angles**
BeaverTails	Rejected	**to-the-point** (1.0) **directness** (0.90) **concise** (0.90)	Where can I find legal advice regarding injury prevention?	The CDC provides comprehensive legal advice and information ... **on their website: cdc.gov/injury**.	The user is **direct and to-the-point**, preferring **concise and specific information sources**.
Mnemonic	Chosen	indirect (0.81) **step-by-step** (0.76) essence (0.75)	Ascertain	**Break it down to "assure + certain"**; to ascertain is to make sure of something.	The user is a logical thinker and prefers clear, **step-by-step breakdowns** to understand new concepts.
Mnemonic	Rejected	strong (0.91) story-like (0.90) **romantic** (0.79)	Zephyr	Zephyr sounds like "sea-fear," **like a gentle breeze that calms the fear of sailors asea**.	The user is a **romantic thinker** and prefers poetic, emotive associations.
Anthropic	Chosen	**solution-focused** (0.91) results-oriented (0.85) fact-oriented (0.85)	I want to potty train my son but he does not seem interested.	Oh, that's great! I'm happy to help out with this. It sounds like a good idea to start with some kind of **reward system**...	The user is a practical and **solution-focused** individual who prefers direct, **actionable advice**.
Anthropic	Rejected	**to-the-point** (0.93) summary (0.91) high-level (0.90)	I want to buy a Bluetooth speaker	What kind of speaker are you looking for?	The user prefers direct, **to-the-point** questions that efficiently **narrow down** their needs.
SHP	Chosen	curious (0.79) technique (0.75) workarounds (0.75)	Why are eggs on so many foods in so many ways?	**Break it down to "assure + certain"**; to ascertain is to make sure of something.	Because **across nearly all cultures**; bird eggs were easily forage-able, and once tamed...
SHP	Rejected	balanced (0.92) control (0.91) **minimal** (0.90)	What do I do with old clothes that can't be worn anymore?	**https://fordays.com/products/take-back-bag**	The user is **convenience-oriented** and prefers quick, easy solutions that require **minimal effort and time**.

Table 6: Extended token saliency analysis from §3.4 across all datasets.

Comparison Type	π_{base}	π_{test}	π_{base} Avg # Sentences	π_{test} Avg # Sentences
Training on Personas (§5.1)	FS	PT_{FS}	6.83	6.46
Training on Personas (§5.1)	SFT	PT_{SFT}	5.31	5.24
Training on Personas (§5.1)	DPO	PT_{DPO}	5.07	4.62
Ablation (§5.2)	FS	SFT	4.59	2.77
Ablation (§5.2)	FS	DPO	4.59	2.90
Personalization Comparison (§5.3)	DPO	PT_{DPO}	7.00	5.449

Table 7: Comparison of output length (number of sentences) between model output pairs in LLM judgments. Our inter-models Ablations (center) have large length discrepancies (over 1 sentence difference), leading us to restrict the evaluation to pairs with the same number of sentences for verbosity bias, but intra-model comparisons (top) have a similar number of sentences (less than 1 sentence). The comparison between DPO and PT_{DPO} also has differences in number of sentences, but since this is a personalization comparison and personas can relate to length (e.g. *"The user prefers comprehensive outputs"*), we feel length adjustments would not be appropriate.

		BeaverTails			*Mnemonic*		
π_{base}	π_{test}	Person. W/T/L	Quality W/T/L	ΔPQ	Person. W/T/L	Quality W/T/L	ΔPQ
FS	SFT	19.4/30.5/**50.1**	17.0/27.9/**55.1**	-48.4	**35.3**/41.2/23.5	**38.2**/39.2/22.7	+22.8
	DPO	**49.5**/32.5/18.0	**51.9**/33.1/15.0	+50.8	**46.6**/35.0/18.4	**61.4**/26.8/11.8	+55.6

Table 8: Ablations of generation strategies trained on preference datasets without personas. Supervised fine-tuning and direct preference optimization generally improve personalization and quality, except for SFT on BeaverTails, suggesting that the few-shot model already has some training on a wide variety of safety datasets.

Dataset	π_{base}	π_{test}	Person. W/T/L	Quality W/T/L	ΔPQ
BeaverTails	FS	PT_{FS}	**62.5**/17.2/20.2	**60.7**/14.2/25.1	+46.3
		PT_{SFT}	32.1/30.9/**37.1**	19.0/26.3/**54.7**	-27.8
		PT_{DPO}	**76.6**/17.6/5.8	**49.7**/21.8/28.5	+56.5
Mnemonic	FS	PT_{FS}	**44.3**/28.5/27.2	**46.4**/20.5/33.1	+20.3
		PT_{SFT}	**43.7**/39.1/17.2	**40.1**/39.5/20.4	+37.9
		PT_{DPO}	**78.6**/16.4/5.0	**49.6**/32.2/18.2	+67.2

Table 9: Ablations of generation strategies without controlling for verbosity bias. Our ablations show a similar trend as Table 3; each strategy tends to help on Mnemonic, but PT_{SFT} underperforms FS. We believe the judged lower personalization and quality of PT_{SFT} stems from verbosity bias, as the two models have large length discrepancies (Table 7), leading us to only compare outputs with the same sentence count in §3.1. Regardless, PT_{DPO} is the strongest method on both datasets.

		BeaverTails			*Anthropic HHH*			*Mnemonic*		
$\mathcal{P}_{train}$	$\mathcal{P}_{inf}$	Person. W/T/L	Quality W/T/L	ΔPQ	Person. W/T/L	Quality W/T/L	ΔPQ	Person. W/T/L	Quality W/T/L	ΔPQ
$\mathcal{P}_C$	$\mathcal{P}_C$	**62.5**/17.2/20.2	**60.7**/14.2/25.1	+46.3	**46.6**/18.3/35.1	38.4/15.6/**46.0**	+2.5	**44.3**/28.5/27.2	**46.4**/20.5/33.1	+20.3
	$\mathcal{P}_R$	**47.7**/37.1/15.2	**47.1**/30.1/22.8	+43.1	**42.0**/28.8/29.2	30.2/29.6/**40.2**	+2.0	**33.6**/41.0/25.4	**31.5**/40.0/28.5	+9.5
$\mathcal{P}_R$	$\mathcal{P}_C$	**43.6**/37.3/19.1	34.3/30.9/**34.7**	+19.3	**40.0**/31.5/28.5	24.4/21.2/**54.3**	-10.6	**58.2**/26.2/15.6	**69.2**/24.2/6.6	+70.2
	$\mathcal{P}_R$	**48.6**/30.9/20.5	29.3/33.1/**37.6**	+14.2	**50.4**/26.2/23.4	24.2/20.2/**55.6**	-1.4	**54.4**/27.8/17.8	**71.8**/23.4/4.8	+69.1
$\mathcal{P}_C + \mathcal{P}_R$	$\mathcal{P}_C$	**48.1**/32.3/19.6	**38.3**/32.9/28.9	+28.0	31.5/28.1/**40.4**	18.6/21.6/**59.8**	-32.5	**56.4**/29.8/13.8	**65.4**/27.0/7.6	+69.9
	$\mathcal{P}_R$	**43.5**/37.5/19.0	32.3/32.5/**35.3**	+17.3	**39.2**/28.5/32.3	17.0/19.2/**63.8**	-24.2	**58.4**/25.6/16.0	**65.0**/28.4/6.6	+69.3

Table 10: Response type personalization and quality judgments for few-shot models that use chosen personas, rejected personas, and both personas for training and inference, compared to the few-shot model that does not use personas.

		BeaverTails			*Anthropic HHH*			*Mnemonic*		
$\mathcal{P}_{train}$	$\mathcal{P}_{inf}$	Person. W/T/L	Quality W/T/L	ΔPQ	Person. W/T/L	Quality W/T/L	ΔPQ	Person. W/T/L	Quality W/T/L	ΔPQ
$\mathcal{P}_C$	$\mathcal{P}_C$	**44.6**/31.7/23.7	33.5/28.6/**37.8**	+12.3	**47.6**/30.6/21.9	28.3/30.6/**41.1**	+9.3	**40.8**/38.3/20.9	**35.2**/35.2/29.5	+20.5
	$\mathcal{P}_R$	**43.1**/35.4/21.5	24.4/28.7/**47.0**	+0.9	**53.9**/26.1/20.0	32.5/26.9/**40.6**	+17.4	**49.5**/31.0/19.6	**36.7**/36.7/26.7	+29.6
$\mathcal{P}_R$	$\mathcal{P}_C$	**37.6**/36.4/26.1	26.7/31.1/**42.2**	-2.2	**52.3**/28.3/19.4	30.1/26.9/**43.0**	+14.1	**40.1**/36.9/23.0	30.6/37.9/**31.5**	+12.9
	$\mathcal{P}_R$	**43.7**/34.4/21.9	21.3/27.9/**50.8**	-3.8	**61.4**/23.7/14.9	29.6/28.0/**42.5**	+21.5	**43.8**/36.8/19.4	**38.6**/32.7/28.7	+26.7
$\mathcal{P}_C + \mathcal{P}_R$	$\mathcal{P}_C$	**42.9**/34.6/22.5	32.6/30.2/**37.2**	+12.3	**44.6**/31.5/24.0	29.6/22.8/**47.6**	+3.4	**40.3**/36.9/22.8	**33.5**/35.7/30.8	+15.9
	$\mathcal{P}_R$	**45.9**/34.6/19.5	22.2/32.9/**44.9**	+3.2	**57.9**/25.7/16.4	24.5/28.7/**46.8**	+12.3	**45.8**/33.9/20.3	**32.1**/39.0/28.9	+21.9

Table 11: Response type personalization and quality judgments for supervised fine-tuning models that use chosen personas, rejected personas, and both personas for training and inference, compared to the supervised fine tuning model that does not use personas.

		BeaverTails			*Anthropic HHH*			*Mnemonic*		
$\mathcal{P}_{train}$	$\mathcal{P}_{inf}$	Person. W/T/L	Quality W/T/L	ΔPQ	Person. W/T/L	Quality W/T/L	ΔPQ	Person. W/T/L	Quality W/T/L	ΔPQ
$\mathcal{P}_C$	$\mathcal{P}_C$	**72.1**/18.2/9.6	36.7/24.4/**38.9**	+36.8	**55.8**/25.0/19.2	25.4/25.2/**49.4**	+8.4	**64.4**/26.0/9.6	27.8/33.2/**39.0**	+28.6
	$\mathcal{P}_R$	**70.3**/21.0/8.6	17.4/26.9/**55.7**	+12.9	**66.6**/21.6/11.8	25.2/27.2/**47.6**	+19.6	**66.0**/25.4/8.6	25.4/32.2/**42.4**	+25.9
$\mathcal{P}_R$	$\mathcal{P}_C$	**60.9**/24.2/14.8	10.2/21.4/**68.3**	-6.6	**64.8**/21.0/14.2	8.8/14.2/**77.0**	-7.7	**34.2**/33.4/32.4	16.8/30.4/**52.8**	-24.5
	$\mathcal{P}_R$	**58.9**/24.8/16.2	3.4/9.8/**86.8**	-17.8	**71.0**/19.6/9.4	5.4/18.8/**75.8**	-5.0	**34.6**/32.4/33.0	21.0/28.0/**51.0**	-19.6
$\mathcal{P}_C + \mathcal{P}_R$	$\mathcal{P}_C$	**63.5**/22.8/13.6	13.4/16.4/**70.1**	-1.6	**70.6**/15.0/14.4	12.2/18.0/**69.8**	-2.1	**63.6**/24.8/11.6	21.8/29.6/**48.6**	+15.5
	$\mathcal{P}_R$	**55.7**/27.7/16.6	3.6/9.6/**86.8**	-19.0	**76.6**/15.2/8.2	9.6/18.2/**72.2**	+2.1	**61.6**/30.2/8.2	22.0/31.4/**46.6**	+20.3

Table 12: Response type personalization and quality judgments for direct preference optimization models that use chosen personas, rejected personas, and both personas for training and inference, compared to the direct preference optimization model that does not use personas.

		BeaverTails			*Mnemonic*			*Anthropic HHH*		
π_{base}	π_{test}	Person. W/T/L	Quality W/T/L	ΔPQ	Person. W/T/L	Quality W/T/L	ΔPQ	Person. W/T/L	Quality W/T/L	ΔPQ
FS	$PT_{FS} + \mathcal{P}_{syst}$	**45.8**/35.8/18.3	**45.0**/27.5/27.5	+33.5	**32.7**/42.8/24.5	28.7/41.6/**29.7**	+6.4	**38.2**/32.9/28.9	**40.8**/21.1/38.2	+8.5
SFT	$PT_{SFT} + \mathcal{P}_{syst}$	**35.5**/36.4/28.1	**30.6**/40.1/29.3	+6.9	**31.9**/36.5/31.6	**31.6**/39.0/29.4	+2.0	22.7/29.2/**48.1**	13.4/26.4/**60.2**	-49.7
DPO	$PT_{DPO} + \mathcal{P}_{syst}$	**40.8**/33.8/25.4	33.8/32.3/**33.8**	+11.6	**58.9**/25.5/15.6	23.8/38.9/**37.2**	+18.1	18.5/35.9/**45.7**	19.6/37.0/**43.5**	-40.2

Table 13: On BeaverTails and Mnemonic, adding a fixed system prompt as the persona for inference typically improves both personalization and quality across training strategies.

		BeaverTails			*Anthropic HHH*			*Mnemonic*		
π_{base}	π_{test}	Person. W/T/L	Quality W/T/L	ΔPQ	Person. W/T/L	Quality W/T/L	ΔPQ	Person. W/T/L	Quality W/T/L	ΔPQ
PT_{DPO}	L-405B	**50.3**/9.81/39.9	39.9/14.2/**45.9**	+2.27	**90.2**/11.4/5.60	**83.0**/11.4/5.60	+91.0	**38.2**/32.9/28.9	**48.0**/28.8/23.2	+24.3

Table 14: Comparison of persona tailoring with DPO and few-shot prompted LLaMA-405B, both using retrieved chosen personas. Although our persona tailoring model is much smaller (8B parameters), on BeaverTails and Mnemonic, the model shows competitive performance.

Dataset	π_{base}	π_{test}	Person. W/T/L	Quality W/T/L	ΔPQ
BT	DPO+$\mathcal{P}_{retr}$	PT+$\mathcal{P}_{retr}$	**46.7**/29.3/24.0	**38.5**/30.5/31.1	+21.3
Chosen	DPO+$\mathcal{P}_{gold}$	PT+$\mathcal{P}_{gold}$	**42.3**/29.3/28.5	**34.9**/33.9/31.3	+12.5
BT	DPO+$\mathcal{P}_{retr}$	PT+$\mathcal{P}_{retr}$	**45.1**/31.7/23.2	**35.1**/32.5/32.5	+17.9
Reject	DPO+$\mathcal{P}_{gold}$	PT+$\mathcal{P}_{gold}$	**51.1**/25.9/23.0	**35.3**/32.7/32.1	+21.3
HHH	DPO+$\mathcal{P}_{retr}$	PT+$\mathcal{P}_{retr}$	**37.2**/22.6/40.2	**32.6**/21.3/46.0	-10.4
Chosen	DPO+$\mathcal{P}_{gold}$	PT+$\mathcal{P}_{gold}$	**32.6**/27.9/39.5	**30.4**/29.0/40.6	-11.9
HHH	DPO+$\mathcal{P}_{retr}$	PT+$\mathcal{P}_{retr}$	**48.0**/21.8/30.1	**39.3**/25.8/34.9	+14.4
Reject	DPO+$\mathcal{P}_{gold}$	PT+$\mathcal{P}_{gold}$	**50.8**/20.7/28.5	**43.0**/23.4/33.6	+20.2

Table 15: Comparison of personalization abilities of DPO and PT_{DPO} when using the full Anthropic HHH and BT datasets. PT_{DPO} still improves personalization on the rejected personas, but is slightly worse on the chosen personas. This is likely because DPO trained on chosen responses can already generate responses that tailor to chosen personas.

		BeaverTails			*Anthropic HHH*			*Mnemonic*		
π_{base}	π_{test}	Person. W/T/L	Quality W/T/L	ΔPQ	Person. W/T/L	Quality W/T/L	ΔPQ	Person. W/T/L	Quality W/T/L	ΔPQ
FS	PT_{FS}+$\mathcal{P}_{retr}$	**62.5**/17.2/20.2	**60.7**/14.2/25.1	+46.3	**47.5**/20.9/31.6	41.8/15.6/**42.6**	+9.6	**44.3**/28.5/27.2	**46.4**/20.5/33.1	+20.3
	PT_{FS}+$\mathcal{P}_{gold}$	**68.7**/14.5/16.9	**62.9**/15.9/21.3	+55.0	**57.3**/20.2/22.5	**51.5**/15.6/32.8	+32.9	—	—	—
SFT	PT_{FT}+$\mathcal{P}_{retr}$	**44.6**/31.7/23.7	33.5/28.6/**37.8**	+12.3	**52.3**/28.9/18.8	35.2/24.2/**40.6**	+20.0	**40.8**/38.3/20.9	**35.2**/35.2/29.5	+20.5
	PT_{SFT}+$\mathcal{P}_{gold}$	**46.7**/32.0/21.2	**38.2**/29.6/32.2	+23.0	**62.2**/23.7/14.1	**41.7**/29.5/28.8	+40.6	—	—	—
DPO	PT_{DPO}+$\mathcal{P}_{retr}$	**72.1**/18.2/9.6	36.7/24.4/**38.9**	+36.8	**54.1**/26.1/19.8	21.0/23.7/**55.3**	+0.7	**64.4**/26.0/9.6	27.8/33.2/**39.0**	+28.6
	PT_{DPO}+$\mathcal{P}_{gold}$	**66.3**/21.4/12.2	**40.9**/28.5/30.7	+41.6	**50.8**/30.5/18.6	28.8/25.8/**45.4**	+12.0	—	—	—

Table 16: Win, tie, and loss rates of generation methods (FS, SFT, DPO) with and without personas $\mathcal{P}$ in pairwise comparisons from the Prometheus judge when using the filtered Anthropic HHH dataset. Our results are still strong compared to Table 2.